WORKED SOLUTION

STEP III 1988, Question 15

Probability and Statistics • Difference equation

STEP III 1988, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q15

The key idea

Condition on the *first* book of the day. Once it is on the shelf, the rest of the day looks exactly like a fresh day with a shorter shelf, so the expected count for a shelf of length n is tied to the counts for lengths $n-1$ and $n-2$. That gives a linear difference equation, and we solve it just like a second-order recurrence: an auxiliary quadratic for the homogeneous part, plus a particular solution for the constant 1 on the right.

Solution

Throughout, let N_n be the number of books put on the shelf on a day when the shelf has length n, so that $M(n) = \mathrm{E}(N_n)$. Books are thick or thin independently of one another, each thick with probability p. We are told there are always more books than the shelf can hold, so the day always ends with some book failing to fit.

Part (i)

With $n = 0$ there is no room at all. The first book, thick or thin, does not fit, so it and every later book go on the trolley. Hence $N_0 = 0$ always, and $M(0) = 0$.

Part (ii)

With $n = 1$, look at the first book.

- If it is thin (probability q) it fits and fills the shelf. The next book needs at least one unit, so it does not fit, and from then on everything goes on the trolley. So $N_1 = 1$.
- If it is thick (probability p) it needs two units and does not fit, so $N_1 = 0$.

Therefore $M(1) = q \cdot 1 + p \cdot 0 = q$.

Part (iii)

Let $n \geq 2$ and again condition on the first book. Whatever it is, it fits, since it needs at most $2 \leq n$ units.

- If it is thin (probability q), it is shelved and $n-1$ units remain.
- If it is thick (probability p), it is shelved and $n-2$ units remain.

> **Ask yourself** Why is the expected number of *further* books shelved exactly $M(n-1)$ in the first case?

After the first book, the later books arrive in order, are thick or thin independently of the first book with the same probabilities, and are placed by the same rule: shelve until one does not fit, then stop. The only thing the first book has changed is the free space, which is now $n-1$ units. So the remainder of the day is a copy of a whole day with a shelf of length $n-1$, and the expected number of further books shelved is $M(n-1)$. Likewise it is $M(n-2)$ after a thick first book. (This is also true when $n-2=0$, by part (i).)

Counting the first book as well, the law of total expectation gives

$$\begin{aligned} M(n) &= q\big(1+M(n-1)\big) + p\big(1+M(n-2)\big) \\ &= (p+q) + qM(n-1) + pM(n-2), \end{aligned}$$

and since $p+q=1$,

$$M(n) - qM(n-1) - pM(n-2) = 1 \qquad (n \geq 2),$$

as required.

The closed form

First the homogeneous equation $M(n) - qM(n-1) - pM(n-2) = 0$. Trying $M(n) = \lambda^n$ gives the auxiliary equation

$$\lambda^2 - q\lambda - p = 0.$$

Since $q = 1-p$, $\lambda = 1$ is a root ($1-q-p=0$), and the product of the roots is $-p$, so

$$\lambda^2 - q\lambda - p = (\lambda - 1)(\lambda + p).$$

So $A(-p)^n$ and the constant B satisfy the homogeneous equation for any A and B.

Because 1 is a root, a constant cannot produce the right-hand side 1, so we try Cn. Substituting,

$$\begin{aligned} Cn - qC(n-1) - pC(n-2) &= C\big[(1-q-p)n + q + 2p\big] \\ &= C(q+2p) = C(1+p). \end{aligned}$$

This equals 1 when $C = \dfrac{1}{1+p}$, which is fine since $1+p > 0$. By linearity, then, $M(n) = A(-p)^n + B + Cn$ satisfies the equation in (iii) for every $n \geq 2$, whatever A and B are.

Now fit the starting values. Part (i) gives

$$M(0) = A + B = 0, \qquad \text{so } B = -A,$$

and part (ii) gives

$$M(1) = -Ap + B + C = -A(1+p) + \frac{1}{1+p} = 1 - p.$$

Hence

$$A(1+p) = \frac{1}{1+p} - (1-p) = \frac{1-(1-p^2)}{1+p} = \frac{p^2}{1+p},$$

so

$$A = \frac{p^2}{(1+p)^2}, \qquad B = -\frac{p^2}{(1+p)^2}, \qquad C = \frac{1}{1+p}.$$

That is,

$$M(n) = \frac{n}{1+p} - \frac{p^2}{(1+p)^2}\big(1 - (-p)^n\big).$$

Since (i), (ii) and (iii) fix $M(0)$, $M(1)$ and then each later value in turn, this is not just a possible solution but the only one.

Checks

- $p = 0$ (all thin): $M(n) = n$, as it must be.
- $p = 1$ (all thick): $M(n) = \frac{n}{2} - \frac{1}{4}\big(1 - (-1)^n\big)$, which is $\frac{n}{2}$ for even n and $\frac{n-1}{2}$ for odd n, i.e. $\lfloor n/2 \rfloor$. Correct.
- Direct from the recurrence with $p = q = \frac{1}{2}$: $M(2) = 1 + \frac{1}{4} = \frac{5}{4}$, and the formula gives $\frac{2}{3} - \frac{1}{9} \cdot \frac{3}{4} = \frac{5}{4}$.
- For large n each book uses on average $1 + p$ units, so we expect about $n/(1+p)$ books; the formula agrees, with a bounded correction from the wasted space at the end of the shelf.

We checked the closed form against exact enumeration of the shelving process for $n \leq 12$ and several values of p, and by simulation; for example, with $p = \frac{1}{2}$ and $n = 5$ both give $M(5) = \frac{103}{32} \approx 3.219$.

Common slips

- Thinking a thin book can still be shelved after a thick one fails to fit. The question says all later books go on the trolley, which is exactly why $M(1) = q$ and not something larger.

- Trying a constant as the particular solution: because $\lambda = 1$ solves the auxiliary equation, a constant is swallowed by the homogeneous part, and you need Cn.

Beyond STEP

Conditioning on the first step and noticing that what remains is a smaller copy of the same problem is called *first-step analysis*. It is the standard way to find expected hitting times and absorption probabilities in Markov chains. Our answer also illustrates the *renewal theorem*: if items of random size are packed into a length n, the number that fit grows like n divided by the mean size. Grimmett and Welsh, *Probability: An Introduction*, covers both difference equations for random walks and an introduction to Markov chains.

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